Comparative Study of Motion Planning Algorithms for Niryo Robotic Arm

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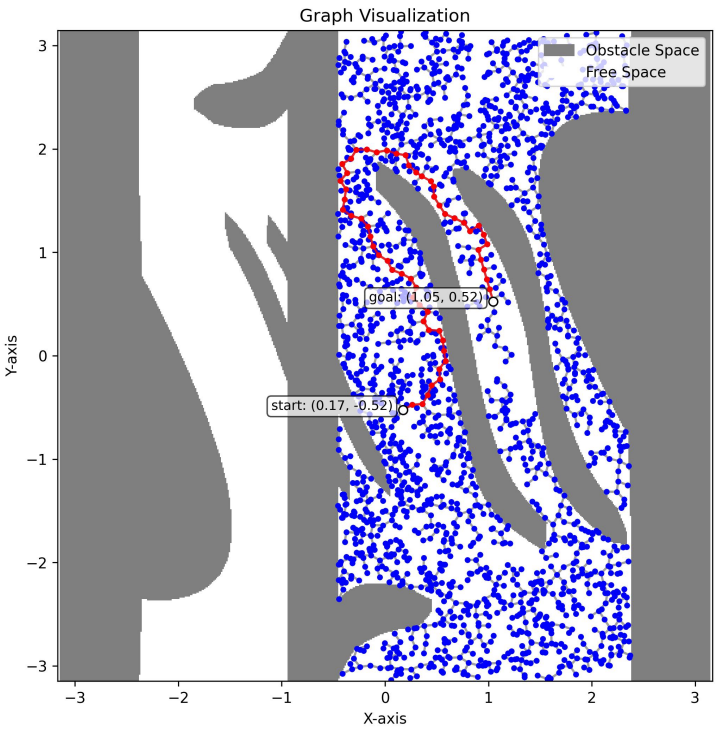
Introduction

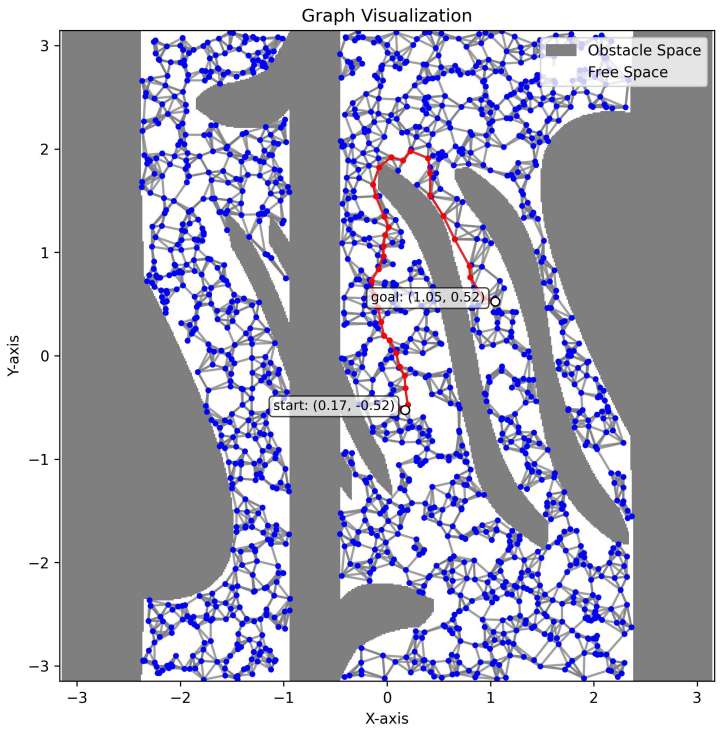
- **Name:** Riku Kawasaki
- **Supervisor:** prof. RNDr. Pavel Surynek, Ph.D.
- **Laboratory:** Robotic Agents Laboratory (RoboAgeLab)
- **Study Department:** Department of Digital Design, FIT CTU in Prague
- **Study Program:** Computer Engineering
- **Thesis Title:** Comparative Study of Motion Planning Algorithms for Niryo Robotic Arm

Background

- Several Motion Planning Algorithms exist
- Resulting motions?
- Computing costs?



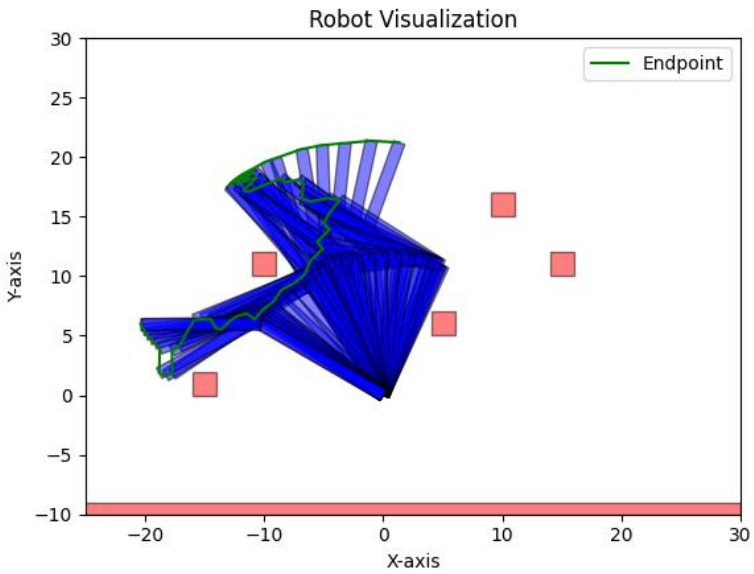
Planning by RRT alg.



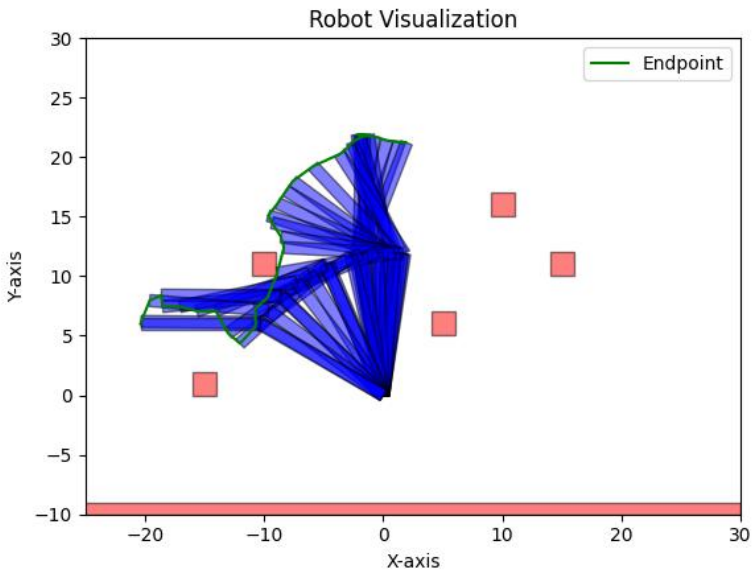
Planning by PRM alg.

Background

- Several Motion Planning Algorithms exist
- Resulting motions?
- Computing costs?



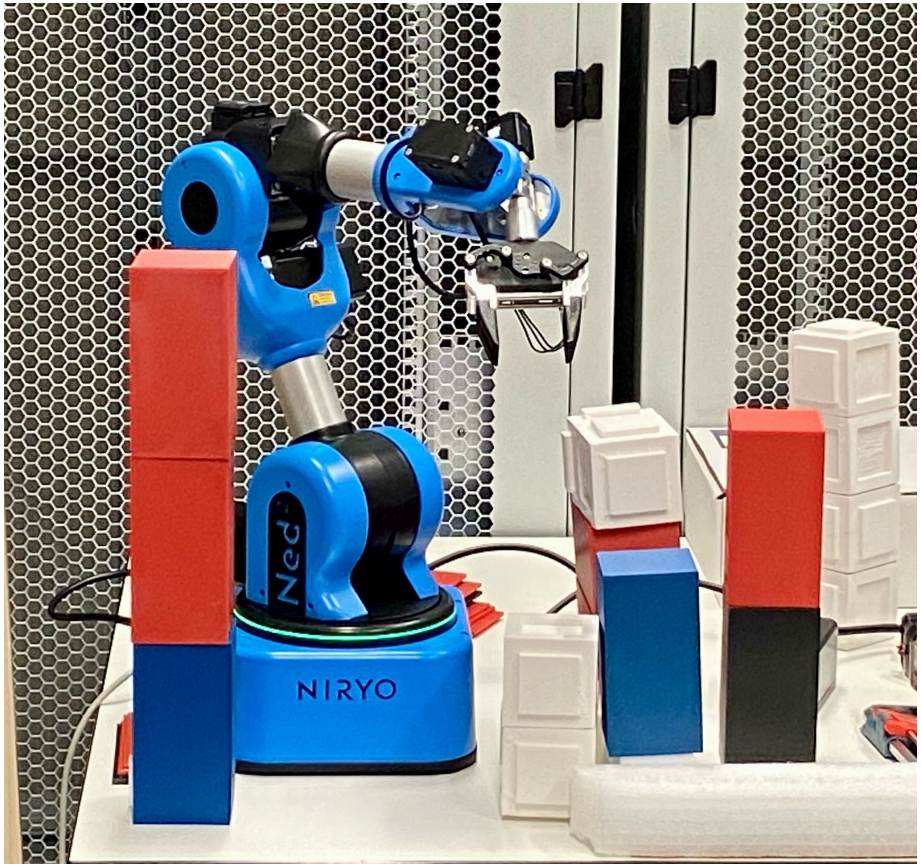
Robot Motion by RRT alg.



Robot Motion by PRM alg.

Aim of Comparative Study

- **Aim:** Evaluate and compare motion planning algorithms in a practical environment using the Niryo Ned2 robot arm.

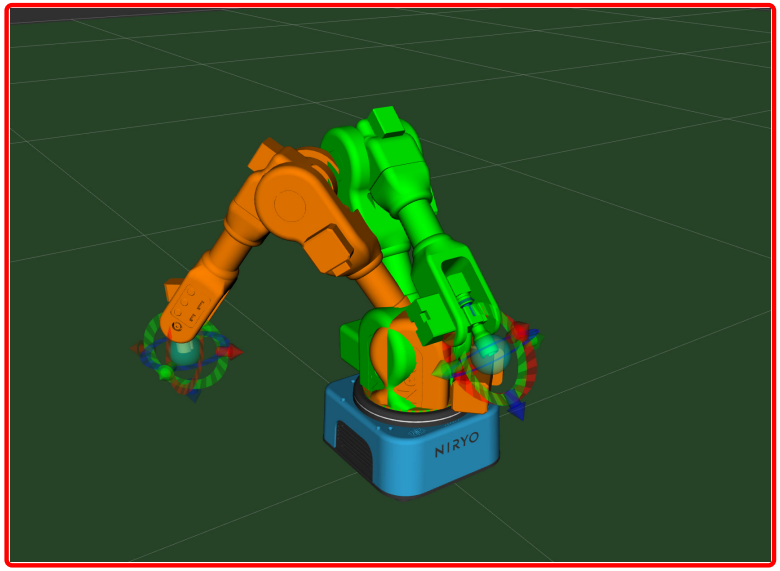


Ned 2 in RoboAgeLab

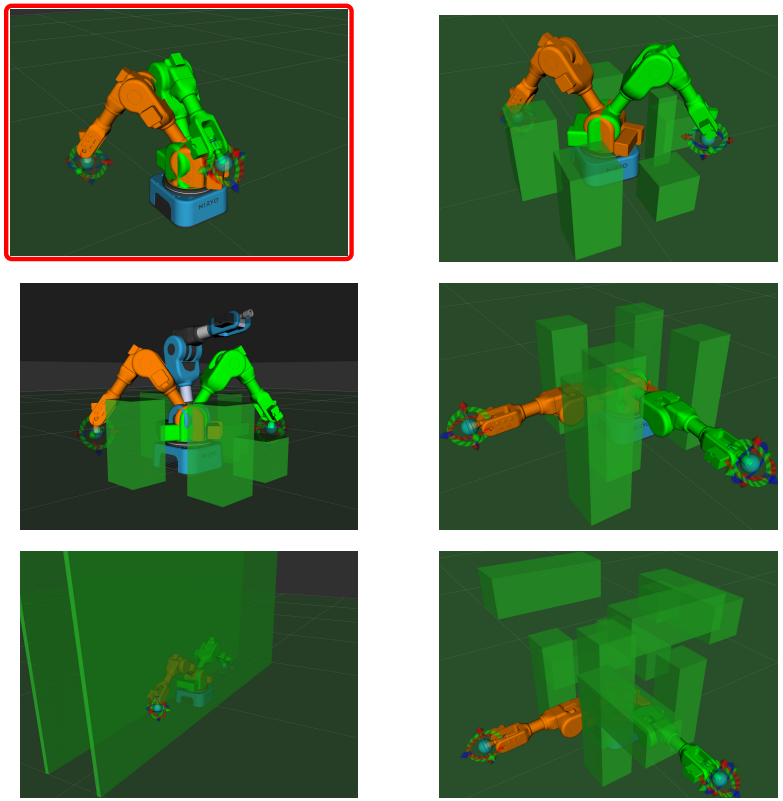
Selected Algorithms

- RRT algorithm
  - RRT (Default)
  - RRTConnect (Bidirectional)
  - RRTstar (for optimized results)
- EST algorithm
  - EST (Default)
  - BiEST (Bidirectional)
- PRM algorithm
  - PRM (Default)
  - pPRM (with reused roadmap)
  - PRMstar (for optimized results)

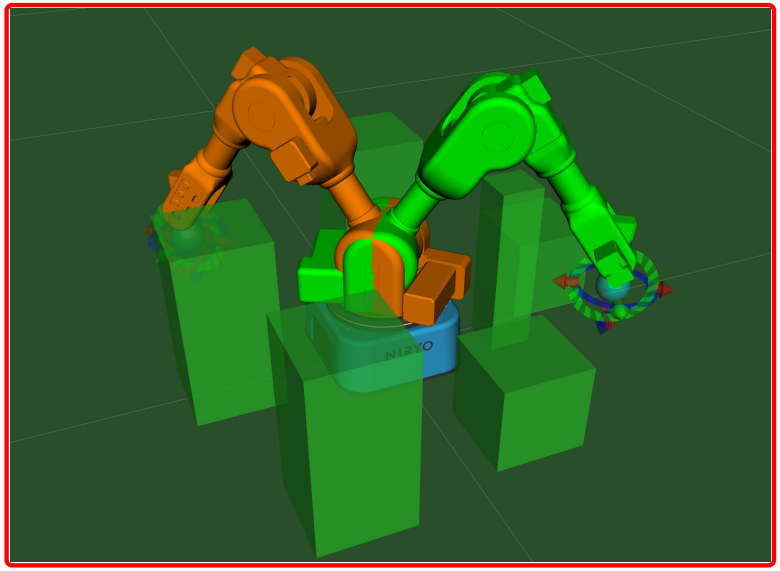
Test Scenes (1/6): No Obstacles



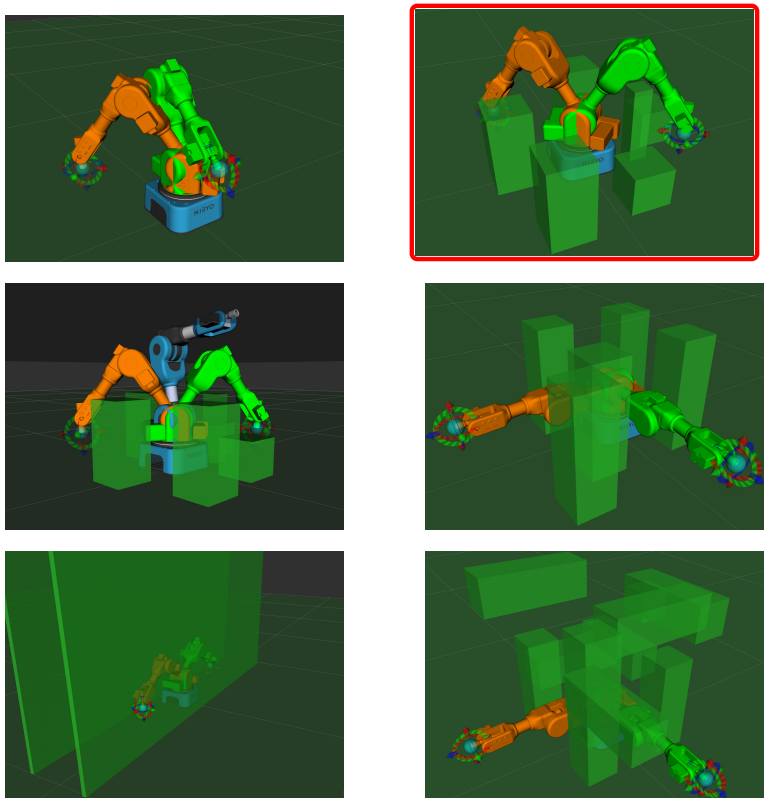
Scene 1: No Obstacles



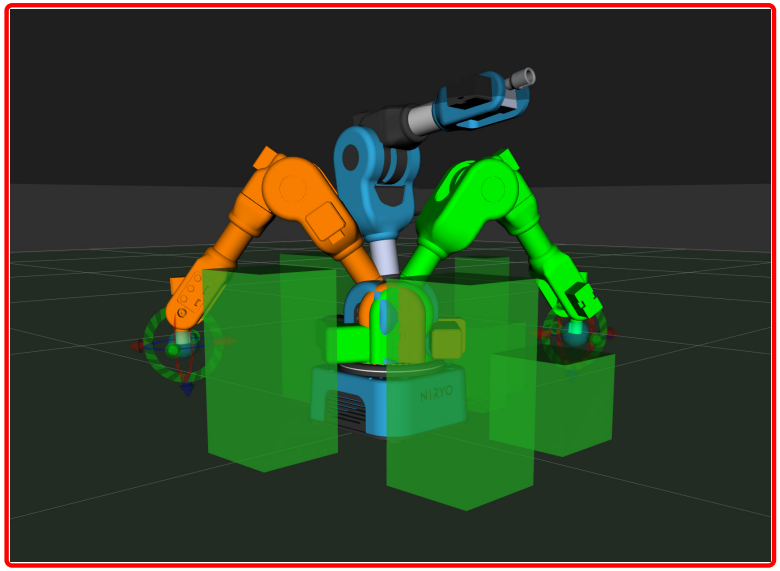
Test Scenes (2/6): Short Obstacles



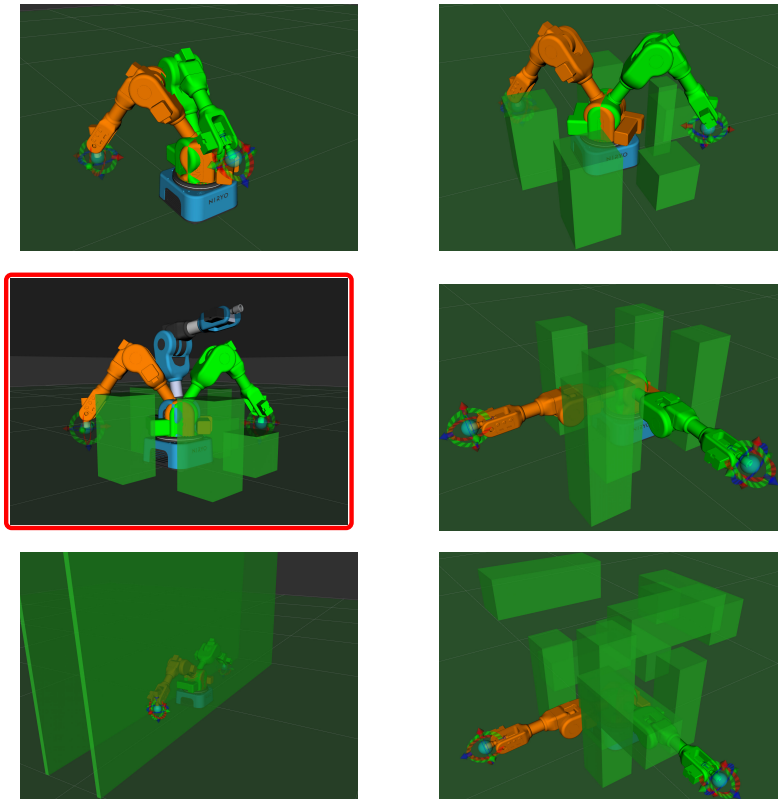
Scene 2: Short Obstacles



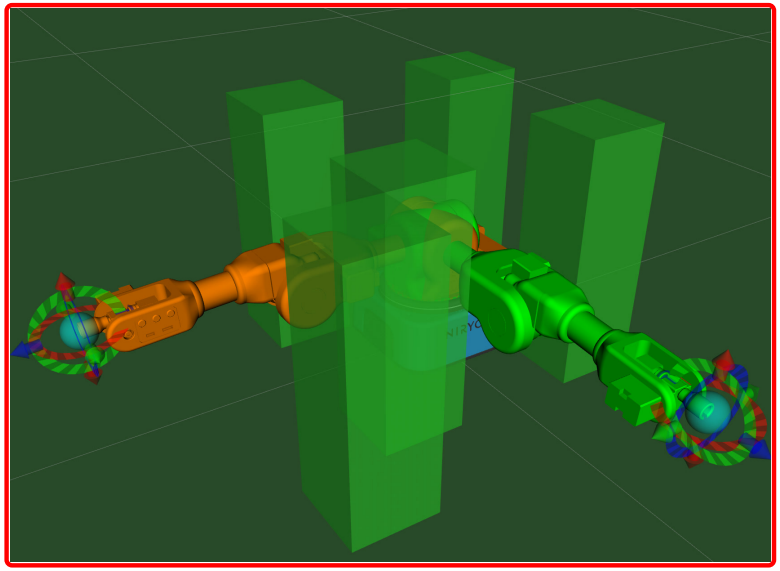
Test Scenes (3/6): Short Obstacles with Ceiling



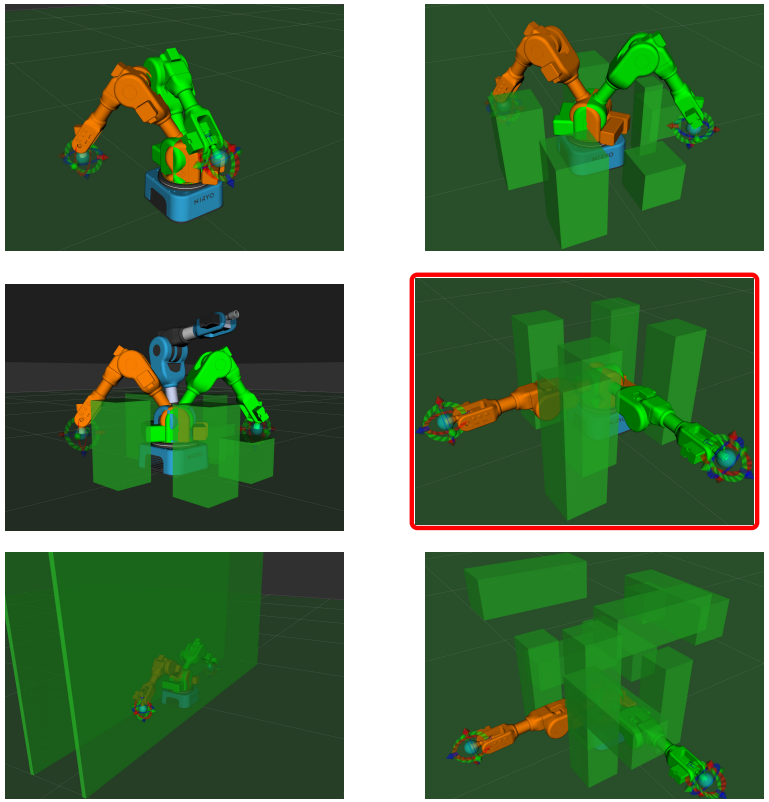
Scene 3: Short Obs. with Ceiling



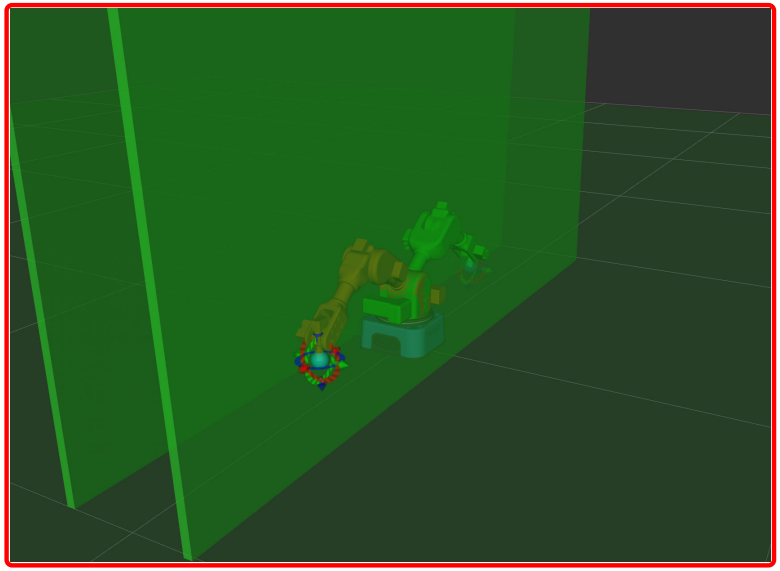
Test Scenes (4/6): Four Columns



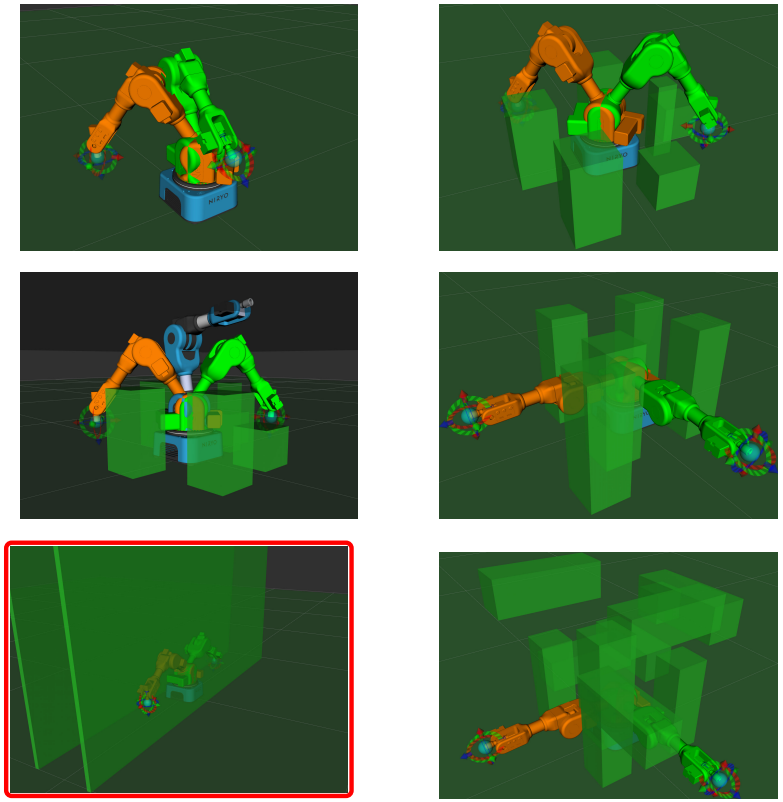
Scene 4: Four Columns



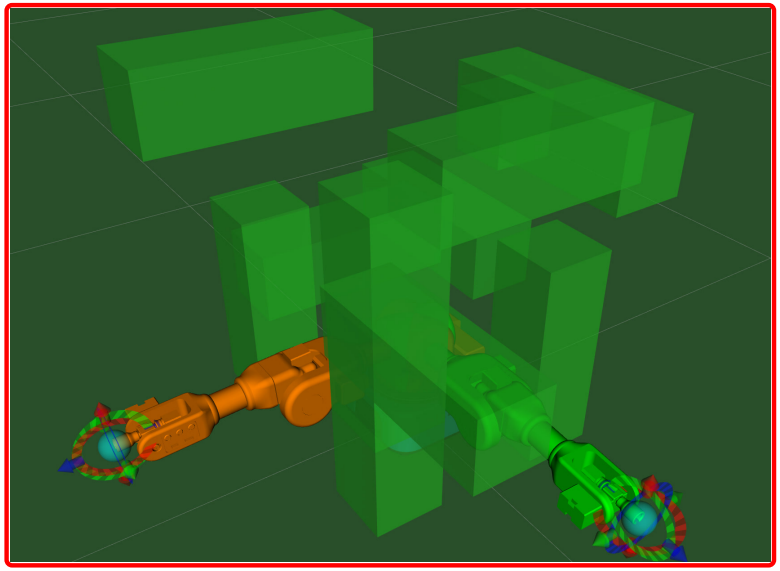
Test Scenes (5/6): Walls



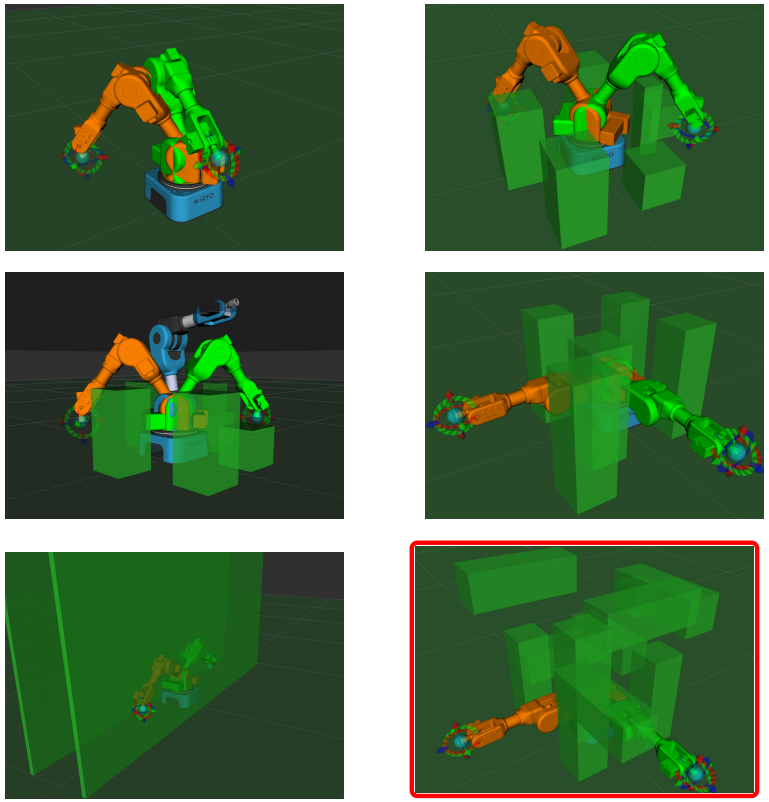
Scene 5: Walls



Test Scenes (6/6): Maze

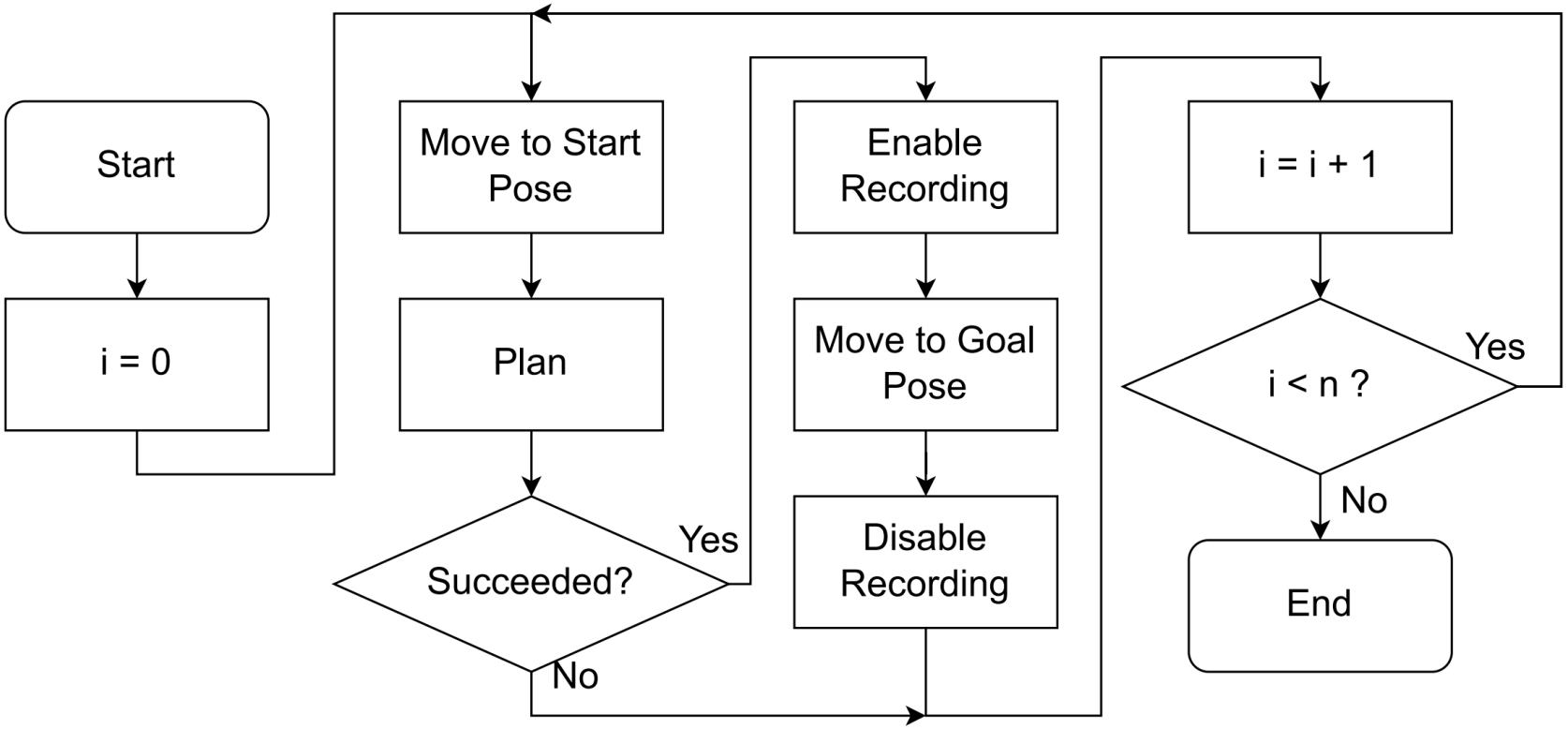


Scene 6: Maze



Automation of Experiment (1/2)

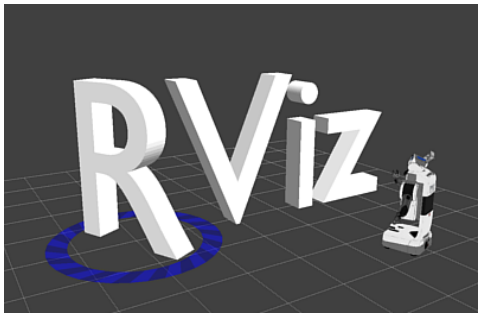
- 8 (alg.) $\times$ 6 (scenes) $\Rightarrow$ **48 (cases)**
- **50 iterations** for each case



Automation of Experiment (2/2)

Used tools:

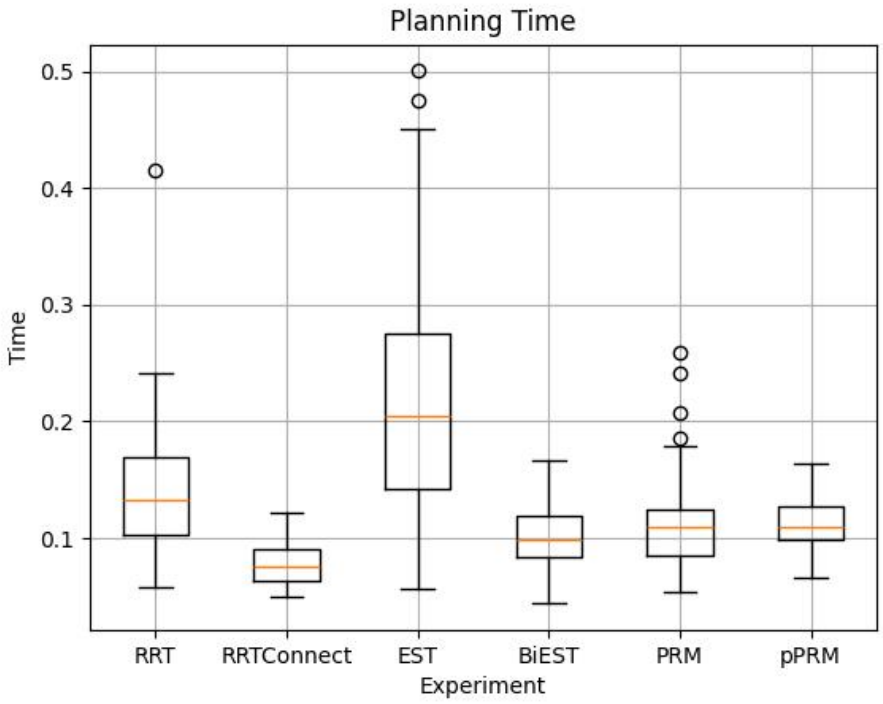
ROS
Robot Operating System



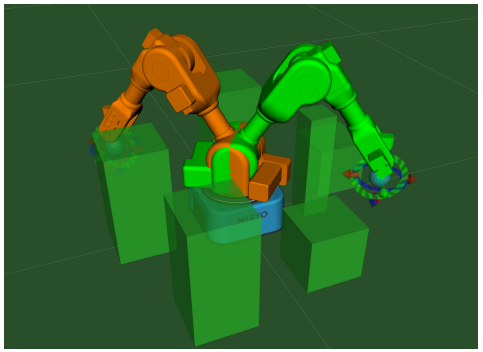
MoveIt

Demo Video: **Click here to watch the video**

Results for Algorithm Efficiency

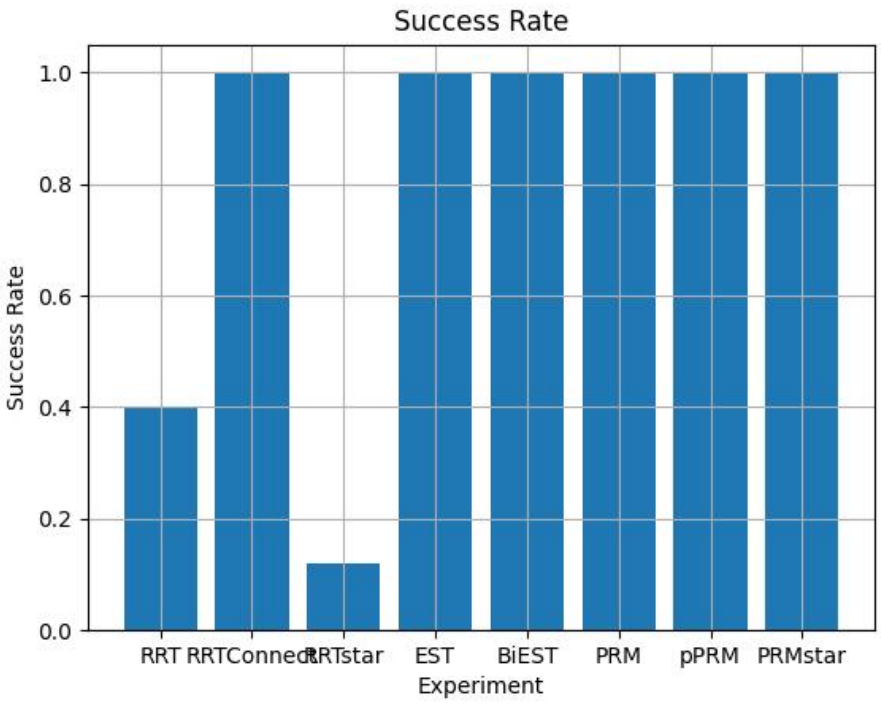


- EST takes a long planning time
- RRTConnect takes the shortest planning time
- Common in test scene 1, 2, 3, 5

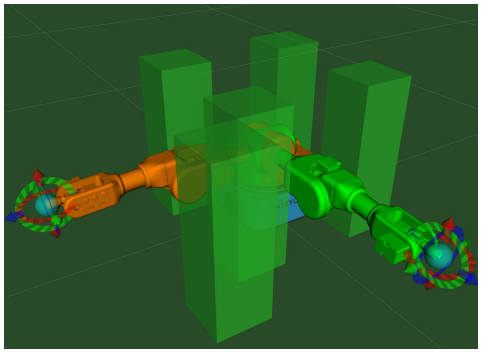


Test Scene 2

Results for Algorithm Efficiency (Exception 1)

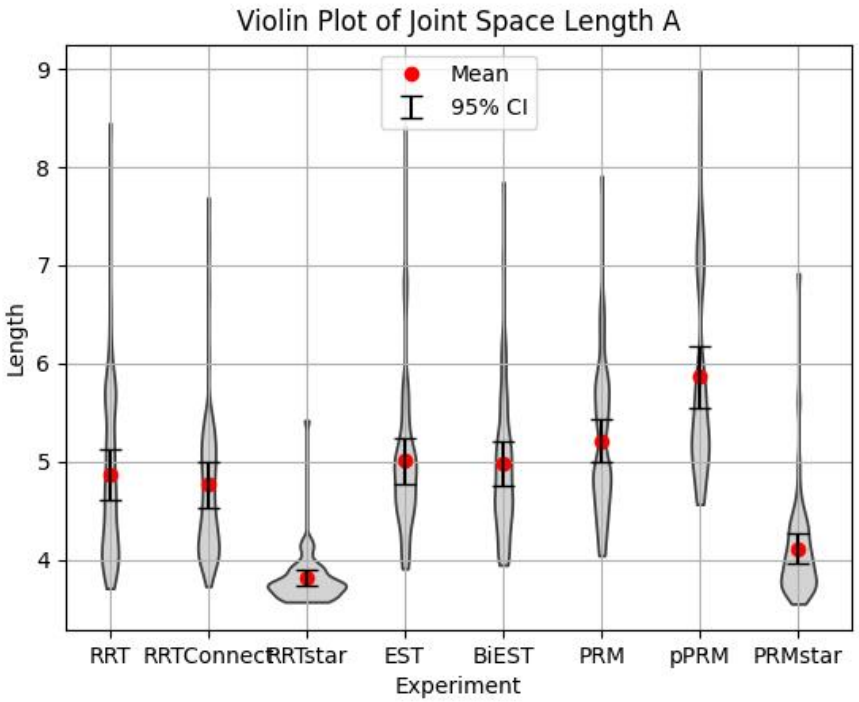


- RRT and RRTstar struggles
- RRTConnect did not get affected

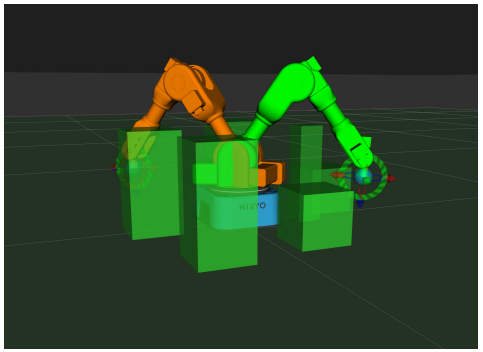


Test Scene 4

Results for Quality of Planned Motions

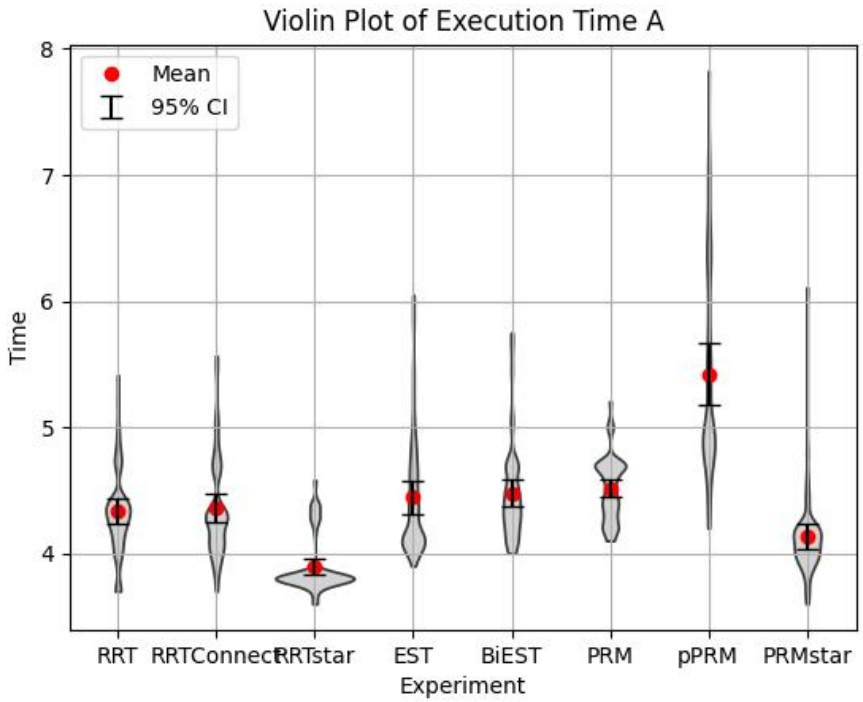


- RRTstar is the best followed by PRMstar
- Similar results in Scene 2, 3, 4, and 5



Test Scene 3

Results for Quality of Planned Motions



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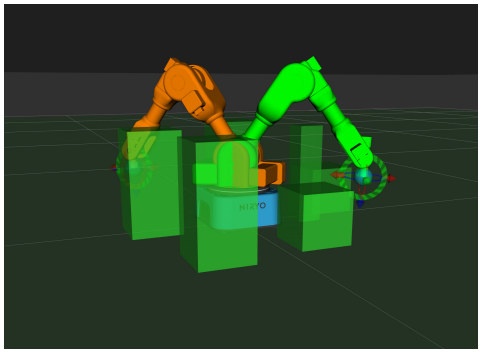
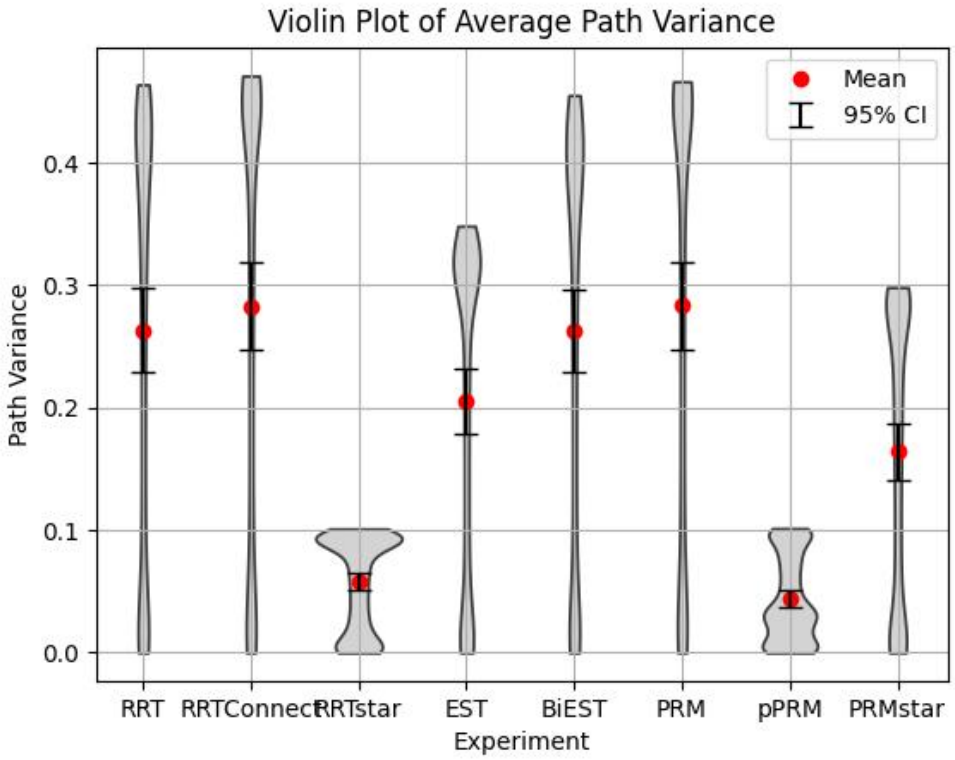
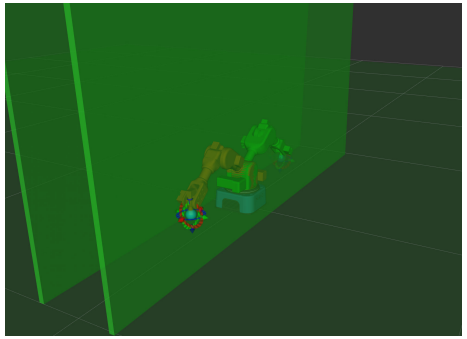


Test Scene 3

Results for Consistency of Motion Planning

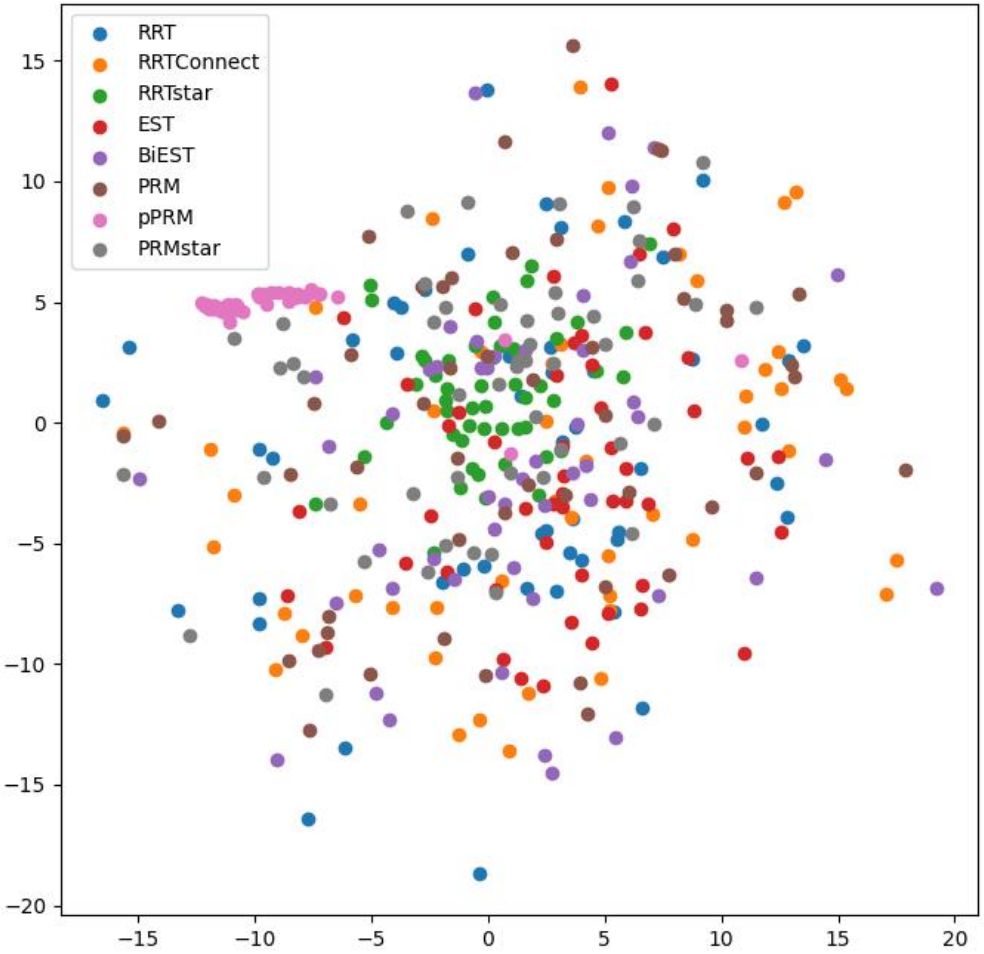


- RRTstar has a small variance
- PRMstar's variance is lower than the most of algorithms, but not as small as RRTstar's

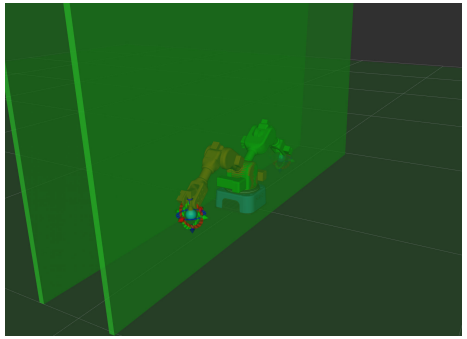


Test Scene 5

Results for Consistency of Motion Planning



2D representations of executed trajectories (left)



Test Scene 5

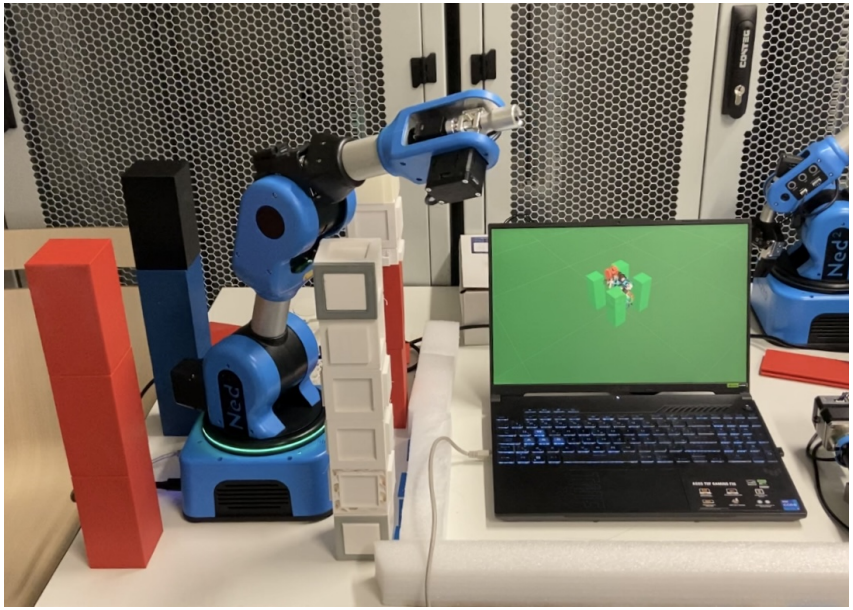
Conclusion

- 8 OMPL planners compared in 6 test scenes using Ned2, ROS, and MoveIt C++ API
- Bidirectional variants (e.g., RRTConnect, BiEST) outperformed their single-tree versions
- Star variants (RRT*, PRM*) produced more optimal paths
- EST had longer planning time than other algorithms
- RRT and RRT* struggled in Test Scene 4

Future Work

- Analyze why Test Scene 4 was challenging for single RRT-based planning
- Explore motion planning algorithms not included in this project
- Extend evaluation to real-time execution environments
- Investigate motion planning in multi-robot scenarios

Thank you for your attention!



Demo Video: **Link to the video**

Riku Kawasaki

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